

ROMA, Sweden

WMO: 025950

Lat: 57.5438N

Lon: 18.4253E

Elev: 32

StdP: 100.94

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.6	-9.8	-15.2	1.0	-12.4	-12.0	1.3	-8.4	11.8	2.7	10.7	2.5	1.0	0	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	26.3	18.7	24.7	17.7	23.2	17.1	20.3	24.0	19.1	22.6	18.1	21.5	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	13.9	21.5	18.0	13.0	20.5	16.8	12.0	19.4	58.5	23.9	54.6	22.6	51.4	21.5	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.1	8.1	-18.4	29.4	4.4	2.2	-21.5	30.9	-24.1	32.2	-26.5	33.4	-29.7	35.0		
(5)			-19.0	21.5	4.4	1.3	-22.2	22.5	-24.8	23.3	-27.3	24.0	-30.5	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-1.3	-1.2	1.1	5.5	10.3	14.2	17.2	16.6	12.4	7.7	4.3	0.9	(6)
(7)		DBStd	7.29	4.17	4.08	3.98	3.04	3.01	2.59	2.49	2.51	2.58	3.22	3.11	4.17	(7)
(8)		HDD10.0	1666	350	314	277	138	33	2	0	0	9	87	172	282	(8)
(9)		HDD18.3	4044	608	548	536	386	251	126	53	65	179	331	422	541	(9)
(10)		CDD10.0	695	0	0	0	2	42	128	223	206	80	14	0	0	(10)
(11)		CDD18.3	31	0	0	0	0	0	2	18	12	0	0	0	0	(11)
(12)		CDH23.3	311	0	0	0	0	1	36	182	92	1	0	0	0	(12)
(13)		CDH26.7	41	0	0	0	0	0	1	28	13	0	0	0	0	(13)
(14)	Wind	WSAvg	3.9	4.5	4.3	4.1	3.8	3.6	3.5	3.2	3.2	3.7	3.8	4.2	4.5	(14)
(15)	Precipitation	PrecAvg	543	46	34	28	26	34	36	56	56	52	53	64	56	(15)
(16)		PrecMax	725	93	51	64	57	72	103	138	116	113	97	137	111	(16)
(17)		PrecMin	431	6	10	5	1	6	0	0	0	14	10	10	16	(17)
(18)		PrecStd	68	19	12	15	15	18	23	40	31	26	22	26	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	7.5	13.1	18.6	22.8	26.1	29.2	28.2	23.2	16.5	11.2	9.4	(19)
(20)			MCWB	6.3	5.6	7.2	10.4	15.7	17.1	19.9	20.9	16.1	13.1	10.2	8.5	(20)
(21)		2%	DB	6.0	6.1	10.6	15.9	20.8	24.3	27.1	25.6	20.4	14.7	10.2	8.1	(21)
(22)			MCWB	5.2	4.9	6.5	9.6	14.3	16.5	19.7	18.5	15.3	12.7	9.3	7.1	(22)
(23)		5%	DB	5.0	4.9	8.1	13.9	18.8	22.6	25.4	23.9	18.8	13.6	9.2	7.0	(23)
(24)			MCWB	4.1	4.0	5.3	8.7	13.1	15.6	18.3	17.8	14.8	12.1	8.3	6.3	(24)
(25)		10%	DB	4.0	3.8	6.2	11.9	16.9	20.9	23.7	22.4	17.3	12.6	8.4	5.9	(25)
(26)			MCWB	3.2	3.0	4.3	7.0	11.9	14.8	17.6	17.1	14.1	11.3	7.7	5.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	6.3	8.2	11.5	17.6	19.1	22.6	21.7	17.3	14.2	10.6	8.8	(27)
(28)			MCDWB	6.9	6.8	11.7	16.4	21.4	23.4	26.2	26.0	20.2	15.3	11.0	9.2	(28)
(29)		2%	WB	5.5	5.2	6.7	10.1	15.5	17.6	20.7	20.1	16.2	13.1	9.4	7.3	(29)
(30)			MCDWB	5.9	5.9	9.7	15.1	18.9	22.8	24.6	23.3	18.7	14.2	9.9	7.9	(30)
(31)		5%	WB	4.3	4.0	5.7	9.0	13.9	16.4	19.6	19.0	15.3	12.3	8.5	6.3	(31)
(32)			MCDWB	4.8	4.8	7.8	13.1	17.4	20.8	23.4	21.8	17.9	13.3	9.0	6.8	(32)
(33)		10%	WB	3.2	3.1	4.6	7.8	12.5	15.4	18.5	18.1	14.5	11.5	7.8	5.4	(33)
(34)			MCDWB	3.8	3.8	6.0	11.0	16.0	19.6	22.2	20.8	16.8	12.5	8.3	5.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	4.6	6.9	9.8	10.4	10.3	9.7	9.6	8.0	6.3	4.0	3.9	(35)
(36)			MCDBR	4.1	4.4	9.9	13.9	13.4	14.5	13.6	12.6	11.1	7.4	4.1	3.1	(36)
(37)			MCWBR	4.0	3.5	6.5	8.0	7.4	7.2	6.5	6.4	6.5	5.5	3.8	3.2	(37)
(38)		5% WB	MCDWB	4.0	4.1	9.0	12.4	11.6	12.3	11.5	10.5	9.7	6.7	3.9	3.2	(38)
(39)			MCWBR	4.0	3.4	6.2	7.5	7.2	6.8	6.2	5.8	6.1	5.3	3.7	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.329	0.343	0.379	0.374	0.365	0.396	0.388	0.357	0.346	0.327	0.292	(40)	
(41)		taud	2.268	2.318	2.361	2.319	2.409	2.473	2.406	2.418	2.479	2.402	2.271	2.200	(41)	
(42)		Ebn at Noon	559	710	808	831	863	875	835	816	790	683	509	442	(42)	
(43)		Edh at Noon	49	72	89	109	106	101	106	98	80	66	49	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.33	0.89	2.41	4.31	5.80	6.30	5.77	4.56	3.03	1.35	0.46	0.23	(44)	
(45)		RadStd	0.05	0.16	0.24	0.44	0.41	0.29	0.46	0.36	0.34	0.21	0.08	0.04	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (30 neighbors)		+0.38	N/A	N/A	N/A	N/A	-109	-141	N/A	N/A	

Nomenclature: See separate page